Power

File: power.kicad_sch

Controller

File: controller.kicad_sch

Sound chip

File: sound_chip.kicad_sch

Audio out

File: audio.kicad_sch

Front panel

File: panel.kicad_sch

How to read this schematic
Each box is a page. Wires end in name tags (labels): any two tags with the same name are connected,
even on different pages. The little arrows/bars marked +5V, +5VA, +3V3 and GND are power connections.
Pages: Power > Controller (Pico 2) > Sound chip (YM2612) > Audio out > Front panel.

Top sheet: double-click a box to open that page

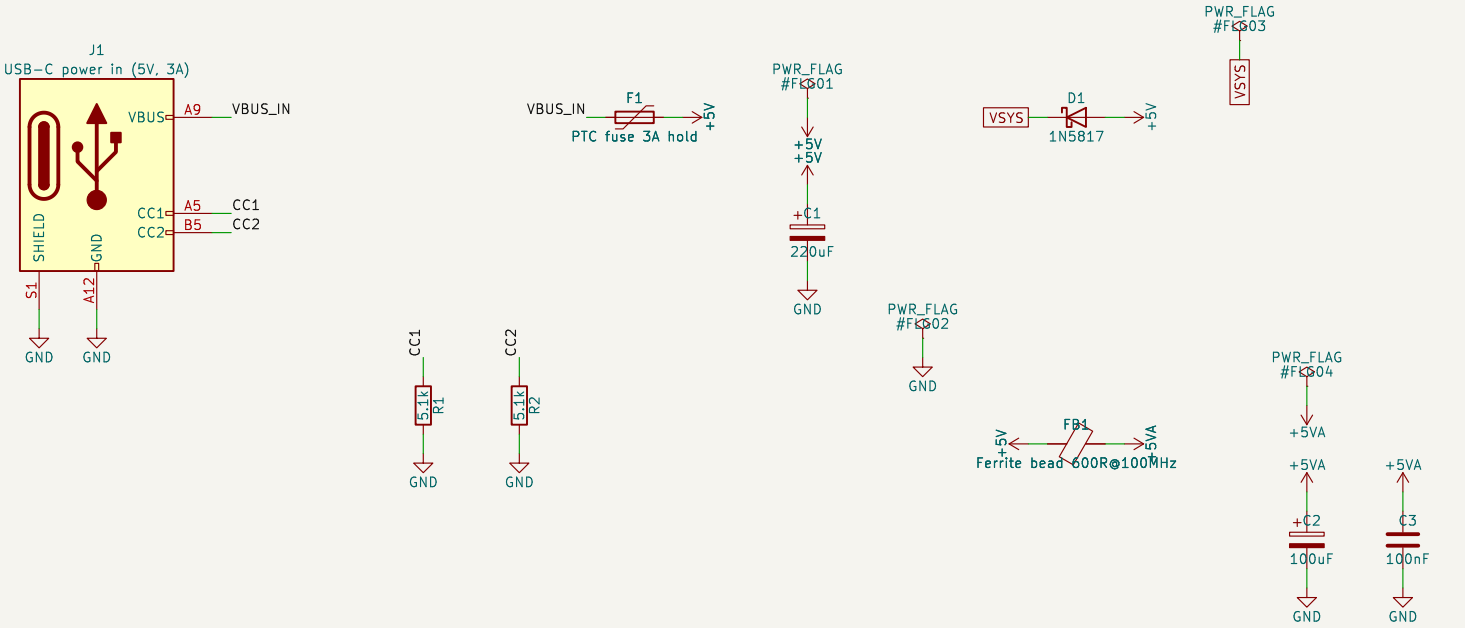
Chip Shrine

Sheet: /
File: chip_shrine.kicad_sch

Title: Chip Shrine: YM2612 player

Size: A4	Date: 2026-10-23	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/6

POWER
USB-C in, 5V. The two 5.1k resistors on CC tell the charger to send 5V.
Use a 5V 3A USB-C supply: the LED grid can pull a lot of current (firmware caps brightness).
+5V = digital 5V and LEDs (straight from USB through the fuse)
+5VA = quiet 5V for the YM2612 and the audio circuit (through a ferrite bead)
VSYS = feeds the Pico through a diode, so plugging USB into the Pico for
programming and the main USB-C at the same time is safe
+3V3 = made by the regulator on the Pico board (see Controller page)
All grounds are one net. On protoboard, run the audio ground and the LED ground
as separate wires back to the USB-C connector (star ground).



YM2612 player proof of concept, draft for review

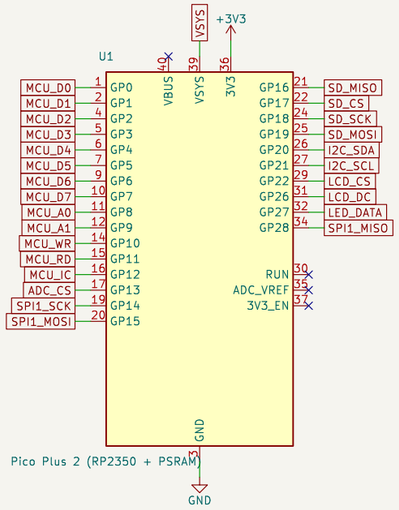
Chip Shrine

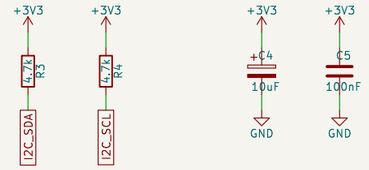
Sheet: /Power/
File: power.kicad_sch

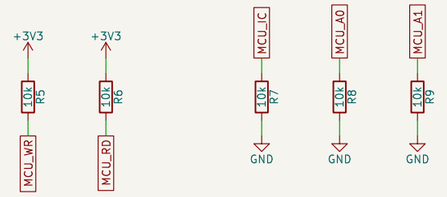
Title: Power

Size: A4	Date: 2026-10-23	**Rev: 0.1**
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CONTROLLER: Raspberry Pi Pico 2 sized board (Pimoroni Pico Plus 2 recommended for its 8 MB PSRAM)
Pin plan
  GP0–GP7   sound chip data bus D0–D7 (driven by PIO, must stay in order)
  GP8, GP9   sound chip A0, A1   GP10 /WR   GP11 /RD (also flips the data buffer)   GP12 /IC (reset)
  GP13   knob ADC chip select
  GP14, GP15, GP28   SPI1 SCK / MOSI / MISO: screen, knob ADC, volume chip
  GP16–GP19 SPI0 to the microSD card (MISO, CS, SCK, MOSI)
  GP20, GP21 I2C0 SDA / SCL to the two button expanders
  GP22, GP26 screen CS and DC   GP27 LED data (grid + indicator LEDs, one chain)
Every Pico GPIO is used. Spare inputs are left on the button expanders.







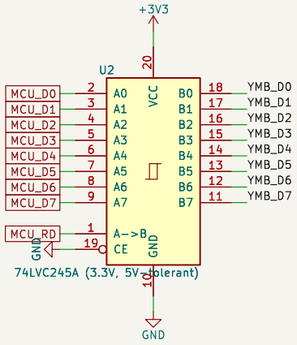
Boot-time pulls: until the firmware starts, these keep the YM2612 in reset and stop the buffers from seeing floating inputs.

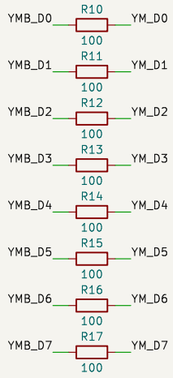
YM2612 player proof of concept, draft for review
Chip Shrine
Sheet: /Controller/
File: controller.kicad_sch

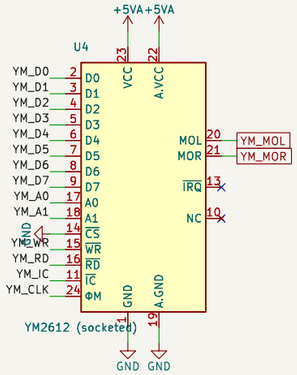
Title: Controller

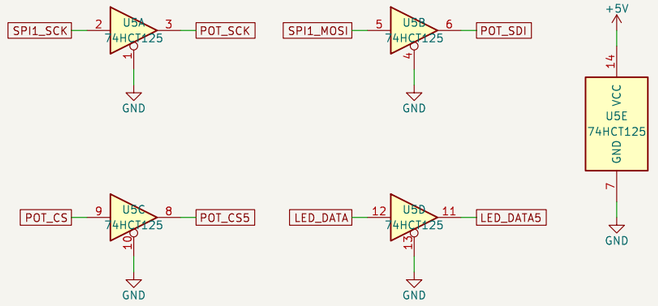
Size: A3	Date: 2026-10-23	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/6

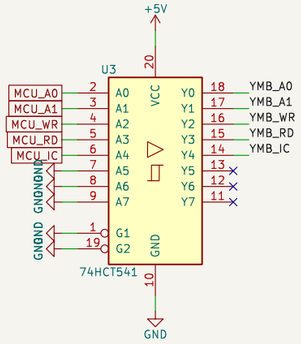
SOUND CHIP: the YM2612 and the level shifters that let the 3.3V Pico talk to the 5V chip
Data bus: 74LVC245A at 3.3V. GP11 (/RD) sets its direction: high = Pico writes to the chip, low = Pico reads the
busy flag. The firmware must switch GP0–GP7 to inputs before pulling /RD low.
Control lines: 74HCT541 at 5V. 100 ohm series resistors on every chip pin protect the YM2612.
/CS is tied low, so the chip listens whenever /WR or /RD is pulsed. /IRQ is not used.
Clock: full-size or half-size oscillator can at 7.670454 MHz (NTSC Genesis), 33 ohm series resistor.
Put the YM2612 in a socket. Keep the clock wire short and away from the audio wiring.

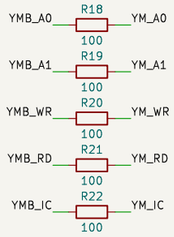


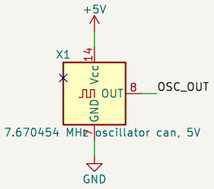


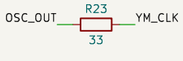


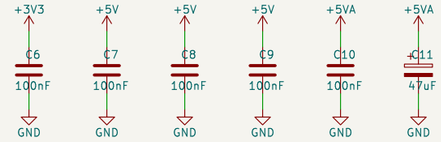








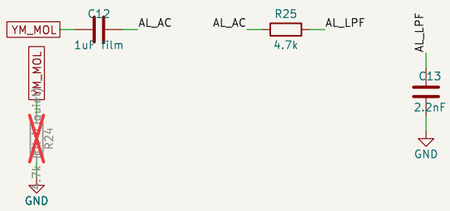


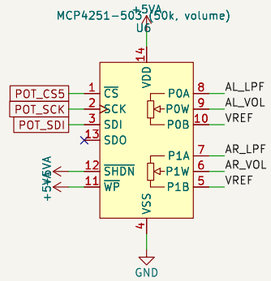


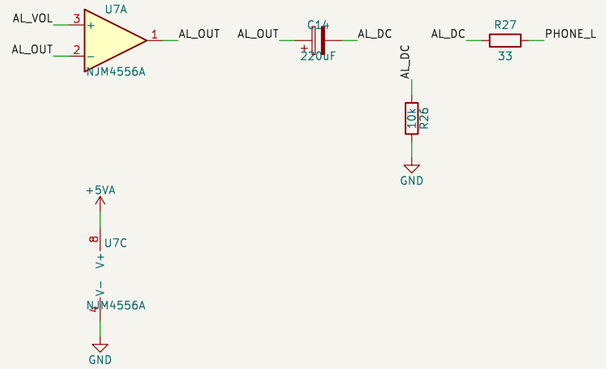
One 100nF capacitor right at each chip's power pin:
U2, U3, X1, U5, U4 = YM2612 (plus 47uF at the YM2612)

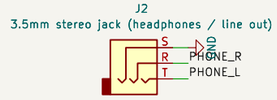
YM2612 player proof of concept, draft for review
Chip Shrine
Sheet: /Sound chip/
File: sound_chip.kicad_sch
Title: Sound chip
Size: A3 | Date: 2026-10-23 | **Rev: 0.1**
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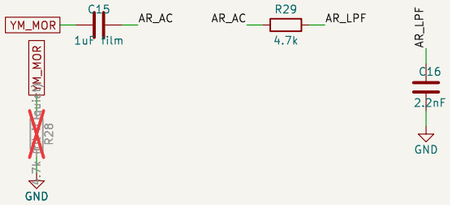
AUDIO OUT: one 3.5mm jack that drives headphones and also works as a line out
Per side: AC-coupling cap > 4.7k + 2.2nF low-pass (about 15 kHz, smooths the DAC steps a little)
> MCP4251 digital pot (volume, set from the encoder over SPI) > NJM4556A buffer (made to drive headphones)
> 220uF output cap > 33 ohm > jack. The 10k resistors bleed the output cap so plugging in doesn't pop.
Everything runs from the quiet +5VA rail around a 2.5V mid-point (VREF).
The 4.7k load resistors on MOL/MOR are marked 'do not fit'. Try without first; fit them if the chip's
output is weak or distorted on the breadboard (milestone M1). Gain is 1x; change it after listening.

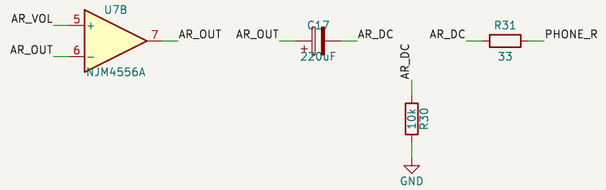


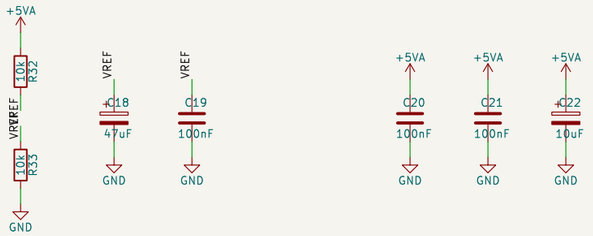








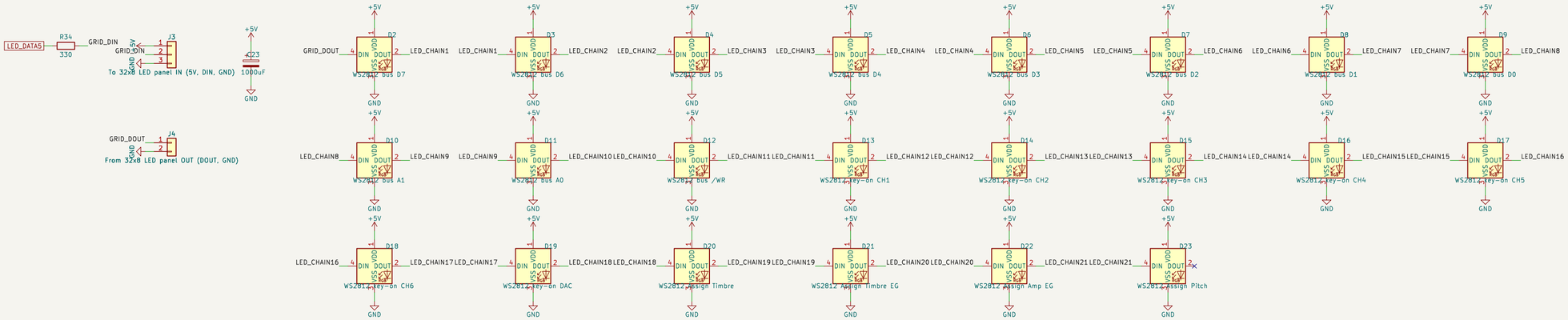




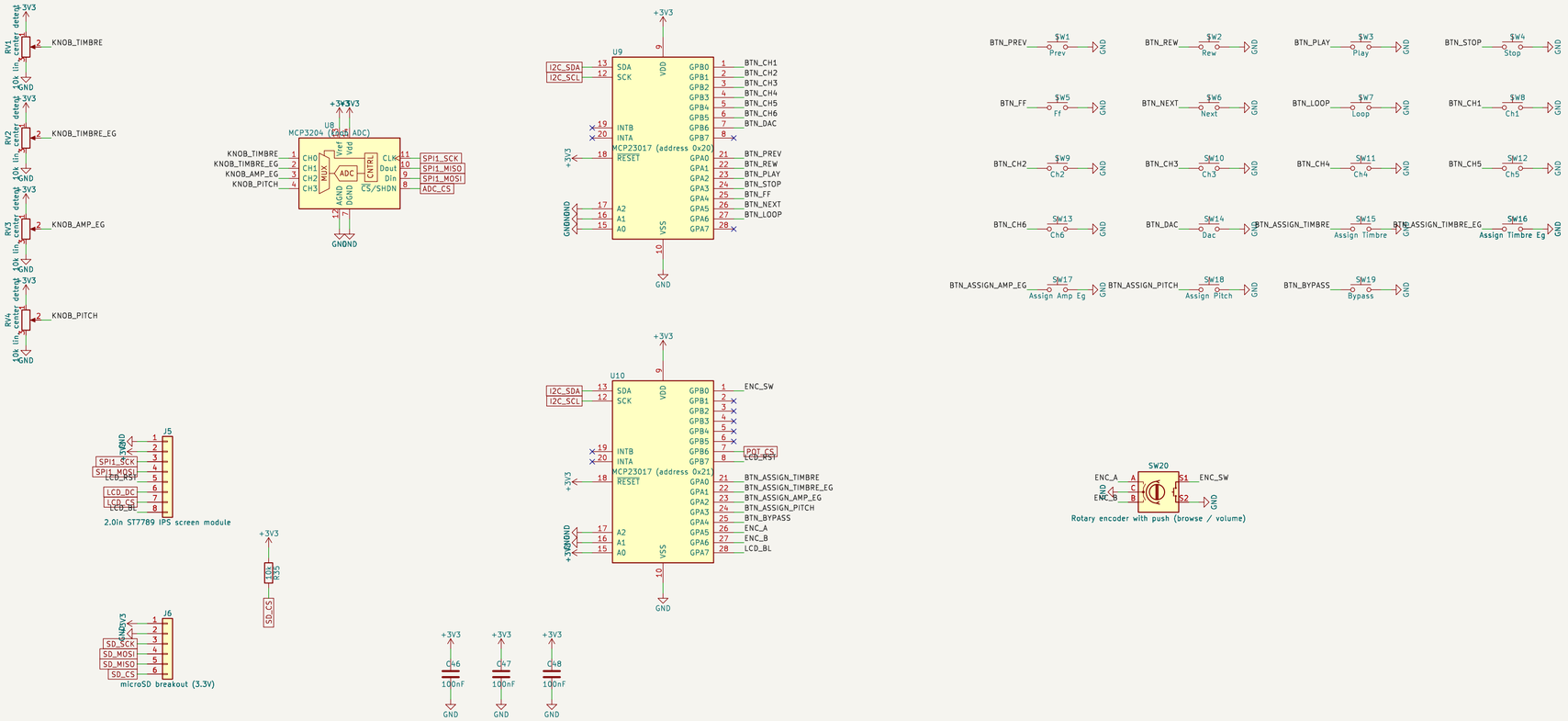
YM2612 player proof of concept, draft for review
Chip Shrine

Sheet: /Audio out/		
File: audio.kicad_sch		
Title: Audio out		
Size: A3	Date: 2026-10-23	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/6

FRONT PANEL: lights, knobs, buttons, screen, card slot
LEDs: one data line from the Pico runs through the 32x8 grid panel first, then on through the 22 indicator LEDs
(11 bus lights, 7 key-on lights, 4 Assign lights). Use 5mm through-hole WS2812 LEDs (e.g. WS2812D-F5) for these;
their pin order differs from the WS2812B drawn here, so check the LED's datasheet when wiring.
Feed the grid's power with thick wires straight from the USB-C connector, not through the protoboard.



KNOBS: 4 potentiometers read by an MCP3204. BUTTONS and the ENCODER: two MCP23017 I2C expanders
with their built-in pull-ups turned on in firmware; every button just shorts its pin to ground.
Expander 0x21 also drives the screen backlight and reset, and the volume chip's select line.



YM2612 player proof of concept, draft for review

Chip Shrine

Sheet: /Front panel/
File: panel.kicad_sch

Title: Front panel

Size: A2	Date: 2026-10-23	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/6